



Signals (Part 1)

ME 350: Design & Manufacturing II

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Some slides adopted from Mike Umbriac, David Remy, Chinedum Okwudire, & Shorya Awtar

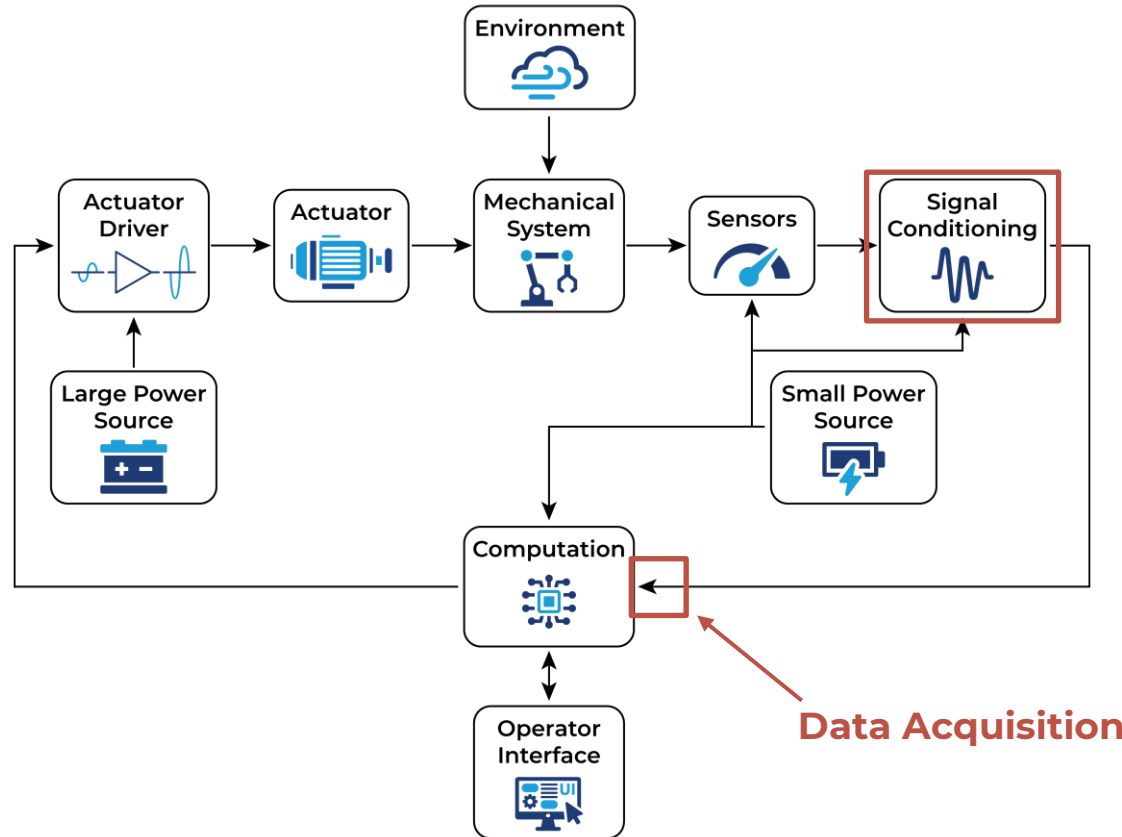
Today's Learning Objectives



By the end of today's class you should...

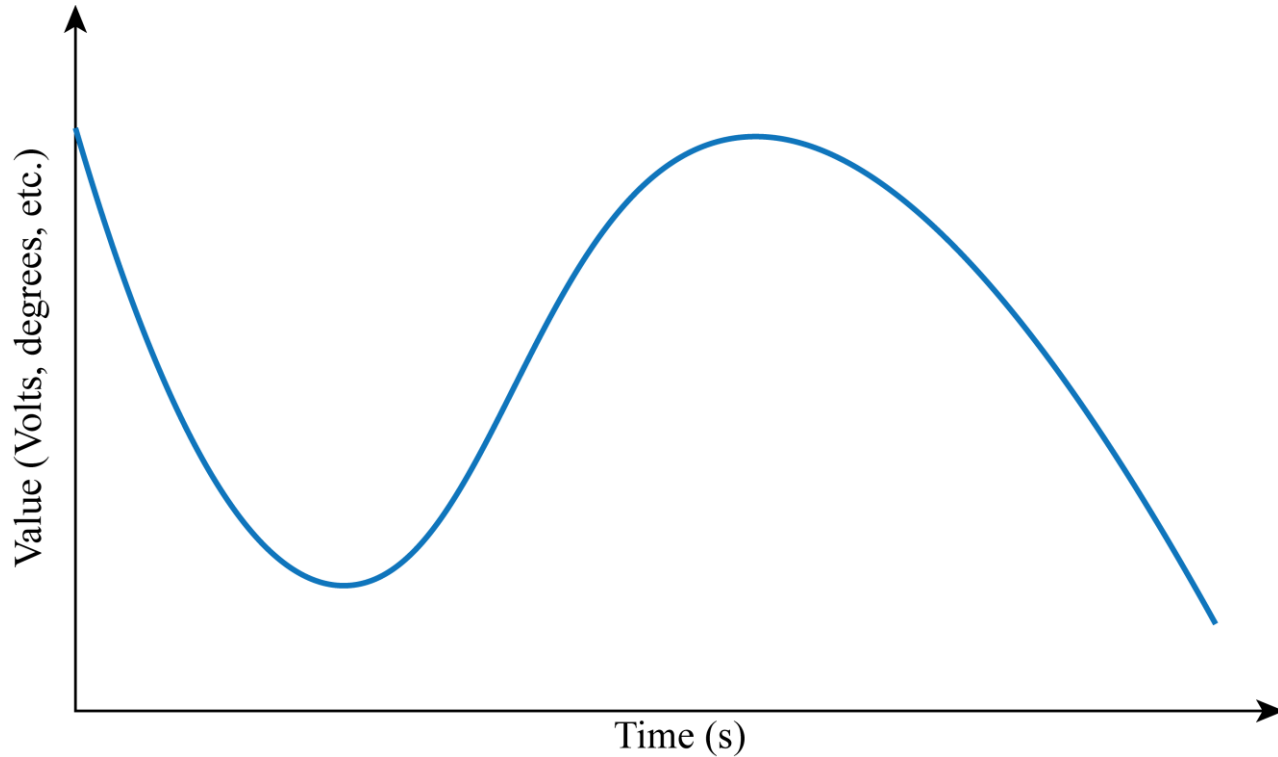
- Be able to understand the ins and outs of various signals
 - Analog Signals
 - Quantized Signals
 - Sampled Signals
 - Digital Signals
- Be able to understand and calculate speed resolution
- Be able to identify the Nyquist frequency and understand its role in aliasing of sampled signals

Context for Today's Lesson



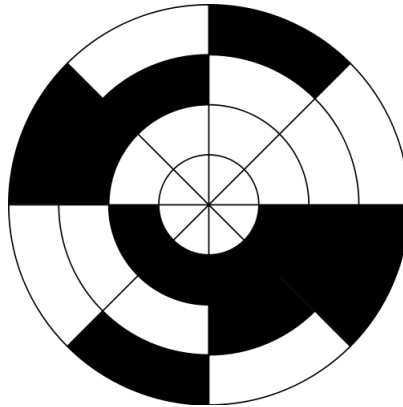
Analog Signal

A fully analog signal is continuous in **time** and **value**



Quantized Signals

- A quantized signal is **discrete in value** but not necessarily **discrete in time**
- Quantization can be caused by a **sensor**
 - Example: An optical encoder with a finite resolution



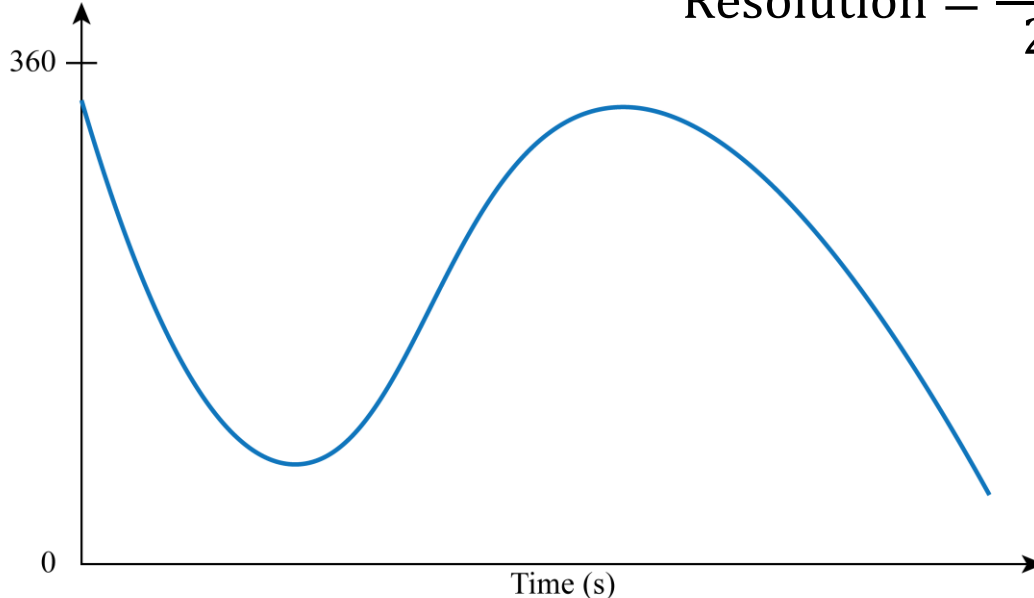
Discrete in value but
continuous in time

- It can also be caused by an **analog to digital convertor (ADC)**
 - We will discuss this more in subsequent lessons

Quantized Signals

Example: Quantize the analog angular position signal below (round down to the lower neighbor) using a 3-bit absolute binary code encoder. **Hint:** To start we need to first calculate the resolution.

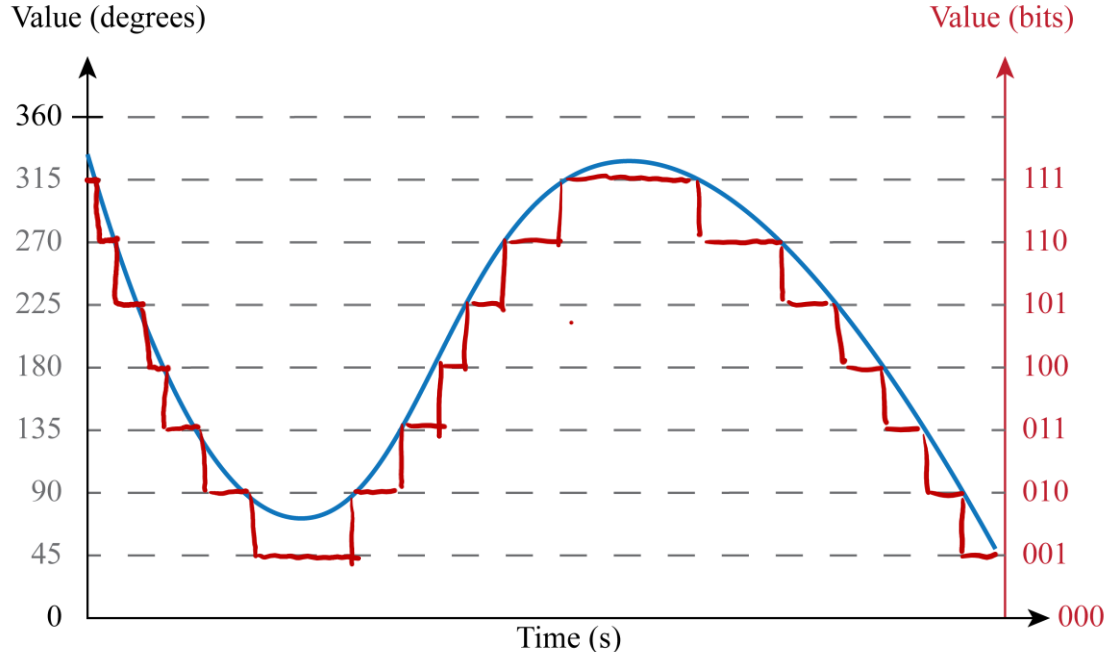
Value (degrees)



$$\text{Resolution} = \frac{360^\circ}{2^n} = \frac{360^\circ}{2^3} = 45^\circ$$

Quantized Signals

Example: Quantize the analog angular position signal below (round down to the lower neighbor) using a 3-bit absolute binary code encoder. **Hint:** To start we need to first calculate the resolution.



Calculating Resolution of a Quantized Signal



Example: You have an analog signal ranging from 0 to 4 V and are trying to represent it using a 2-bit ADC. Can you do this with a resolution of 1 V?

How many discrete values do we get from 2-bits?

$$2^2 = 4$$

2-Bit Binary	00	01	10	11
Resolution of $\frac{4}{2^2} = 1 \text{ V}$	0	1	2	3
Resolution of $\frac{4}{2^2-1} = 1.33 \text{ V}$	0	1.33	2.66	4

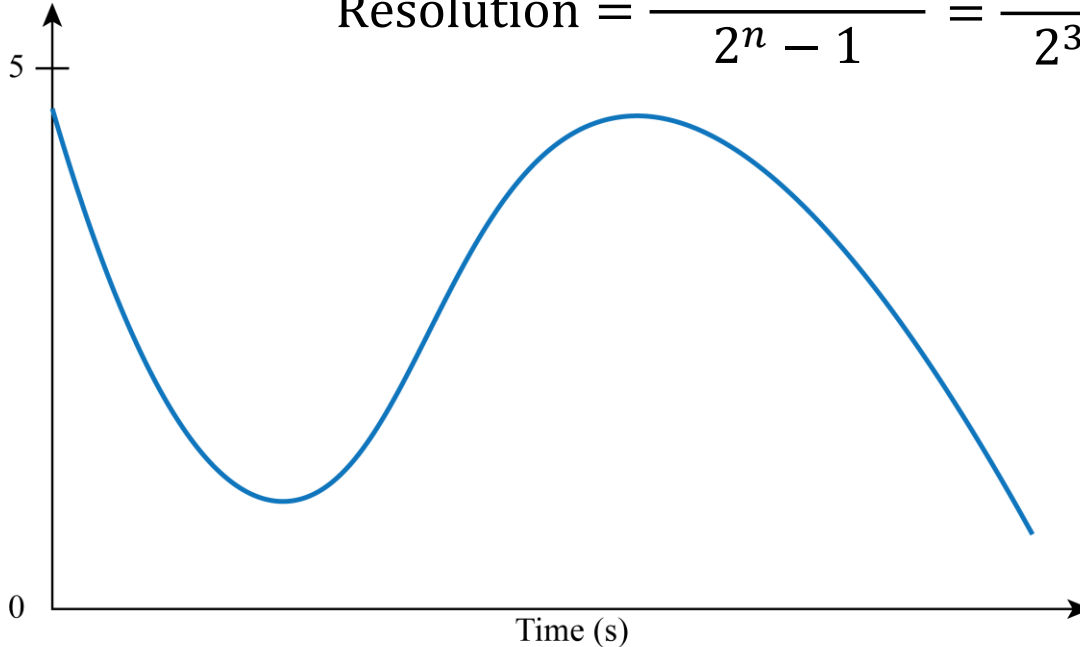
4 is missing!



Quantized Signals

Example: Quantize the analog signal below (to the nearest neighbor) using a 3-bit ADC. **Hint:** To start we need to first calculate the resolution.

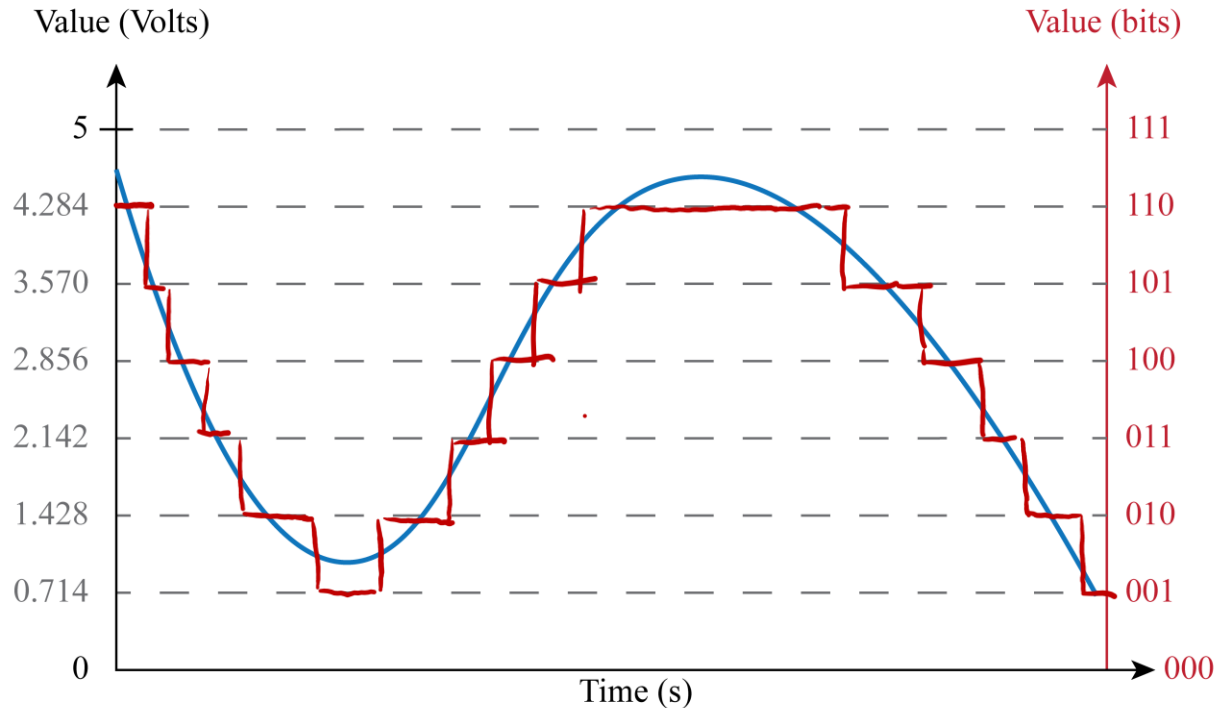
Value (Volts)



$$\text{Resolution} = \frac{V_{max} - V_{min}}{2^n - 1} = \frac{5 \text{ V} - 0 \text{ V}}{2^3 - 1} = 0.714 \text{ V}$$

Quantized Signals

Example: Quantize the analog signal below (to the nearest neighbor) using a 3-bit ADC. **Hint:** To start we need to first calculate the resolution.



Calculating Resolution of a Quantized Signal



n = # of bits of ADC

- When both extremes of the analog signal **do not need** to be represented in the quantized signal

$$\text{Resolution} = \frac{\text{Range}}{2^n}$$

- When both extremes of an analog signal **do need** to be represented by the quantized signal

$$\text{Resolution} = \frac{\text{Range}}{2^n - 1}$$

Sampled Signal

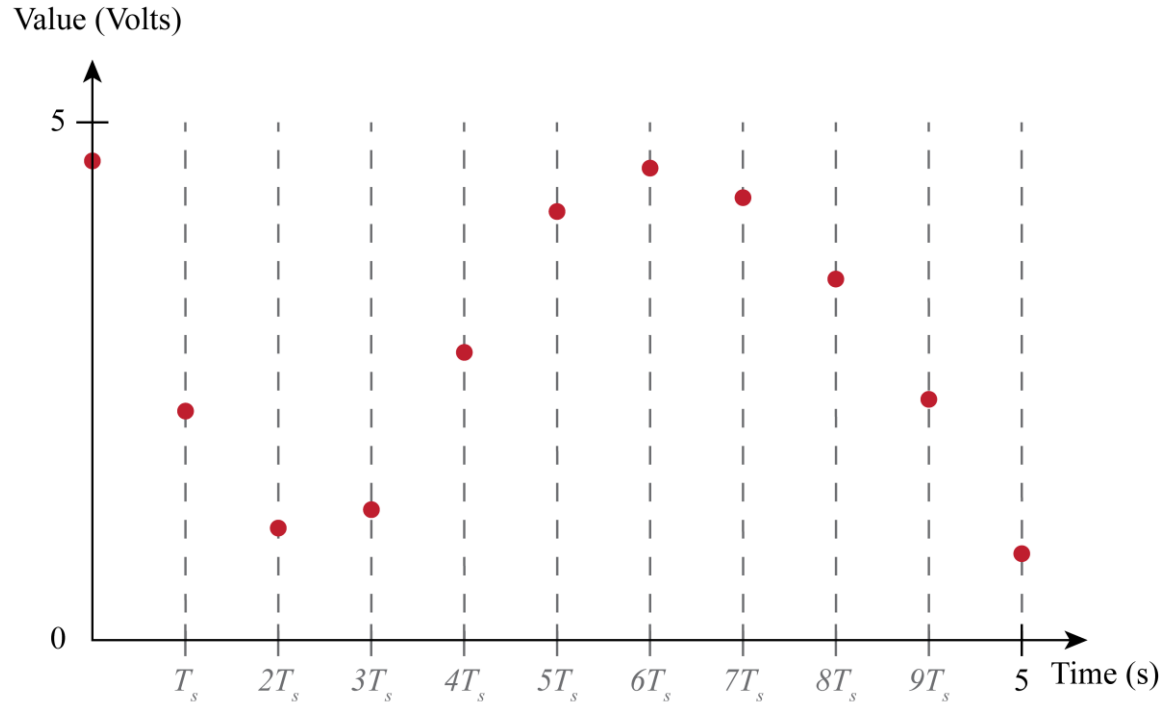


- A sampled signal is **discrete in time** but not necessarily **discrete in value**
- In mechatronic systems, sampling is usually done at **fixed** time intervals (T_s)
- The sampling **frequency/rate** is given by:

$$f_s = \frac{1}{T_s}$$

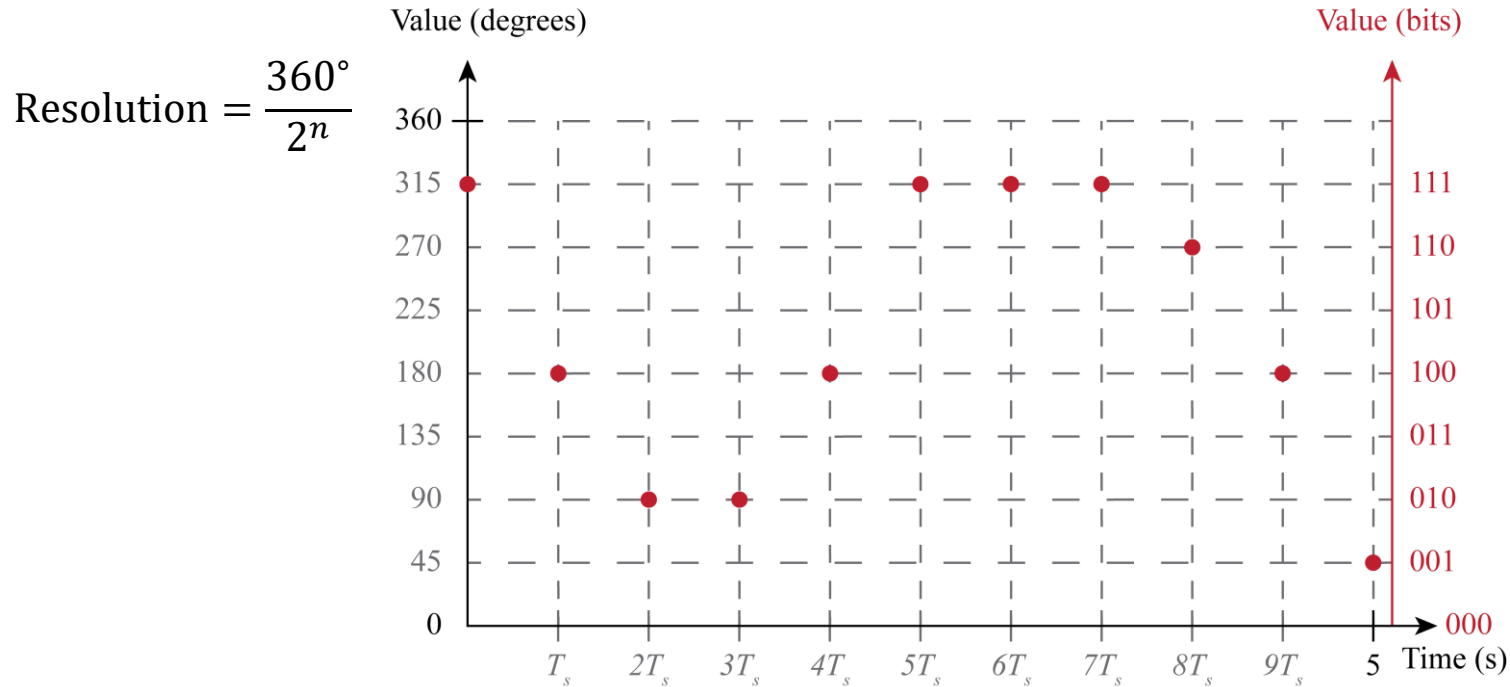
Sampled Signal

Consider the analog signal and let us sample it using a sample interval $T_s = 0.5$ s



Digital Signal

- A (fully) digital signal is discrete in time and value
- Consider the signal below using a 3-bit ADC (nearest neighbor) with $T_s = 0.5$ s



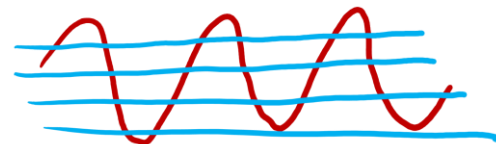
Influence of Sampling & Quantization



Example: Let's change up the quantization for the UM Fight Song



Original Recording (32 bits)



8 bits



4 bits

Music with fewer discrete values **gets “noisy” (i.e. lots of static)**

Effects on Speed Resolution



- Some signals, such as speed, are often calculated by **numerical differentiation** of a sampled and quantized position signal
- When a signal is calculated via numerical differentiation, the resolution is

$$\text{Speed resolution} = \frac{\text{Position Resolution}}{T_s}$$

- This is equivalent to the smallest detectable change in speed that the encoder can measure
- Note that sampling **too fast** (having a smaller T_s) can **worsen** your speed resolution (i.e. make it larger)

Speed Resolution Example

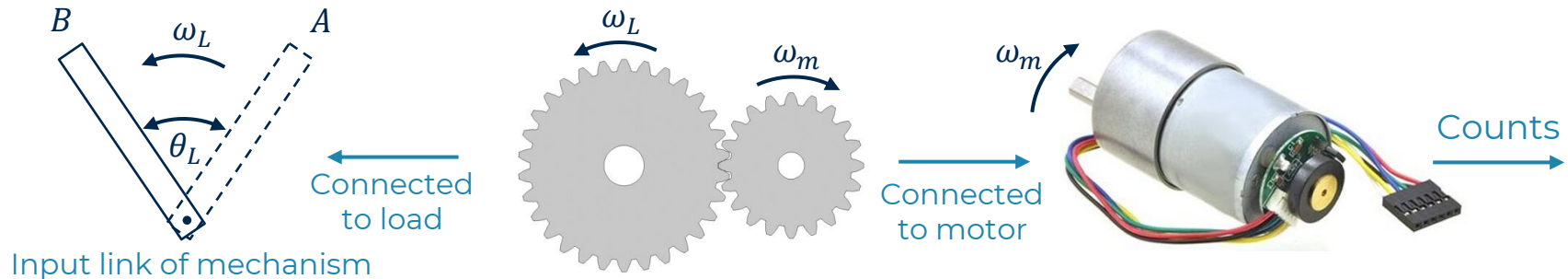
Given:

- A gearmotor with 30:1 internal gear ratio, driving the input link of a mechanism through a gear set with 3:2 speed reduction (i.e. $N = \frac{\omega_m}{\omega_L} = \frac{3}{2}$)
- Encoder has a base resolution of 16 counts per revolution (quadrature decoding) and is attached to the motor shaft
- Sampling interval of $T_s = 5$ ms

Find:

- **The speed resolution** in degrees/s of the input link

$$\text{Speed Res} = \frac{\text{Pos Res}}{T_s}$$



Speed Resolution Example

Given:

- A gearmotor with 30:1 internal gear ratio, driving the input link of a mechanism through a gear set with 3:2 speed reduction (i.e. $N = \frac{\omega_m}{\omega_L} = \frac{3}{2}$)
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Find:

- **The speed resolution** in degrees/s of the input link

$$\text{Speed Res} = \frac{\text{Pos Res}}{T_s}$$

$$\text{Pos Res} = \frac{360^\circ}{(16 \times 4)} \left(\frac{1}{30} \right) \left(\frac{2}{3} \right) = \frac{0.125^\circ}{\text{count}}$$

$$\text{Speed Res} = \frac{0.125^\circ/\text{count}}{0.005\text{ s}} = \boxed{25^\circ/\text{s}}$$

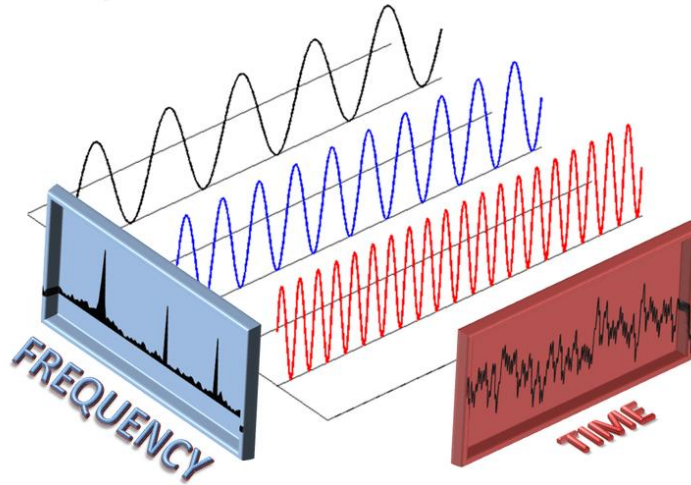
How Fast Should We Sample?



- A signal that varies **quickly** needs to be sampled **faster** than a signal that varies slowly
- If properly sampled, we should be able to closely **reconstruct** the original analog signal
 - Example, when we play an audio file, it should ideally sound like the original music being recorded
- We have to choose the sampling time T_s (units of seconds) or the sampling frequency $f_s = 1/T_s$ (units of Hertz = cycles/second) accordingly

Fourier Transform

- Under certain conditions, any signal can be written as a sum of **sine waves** with different frequencies and amplitudes

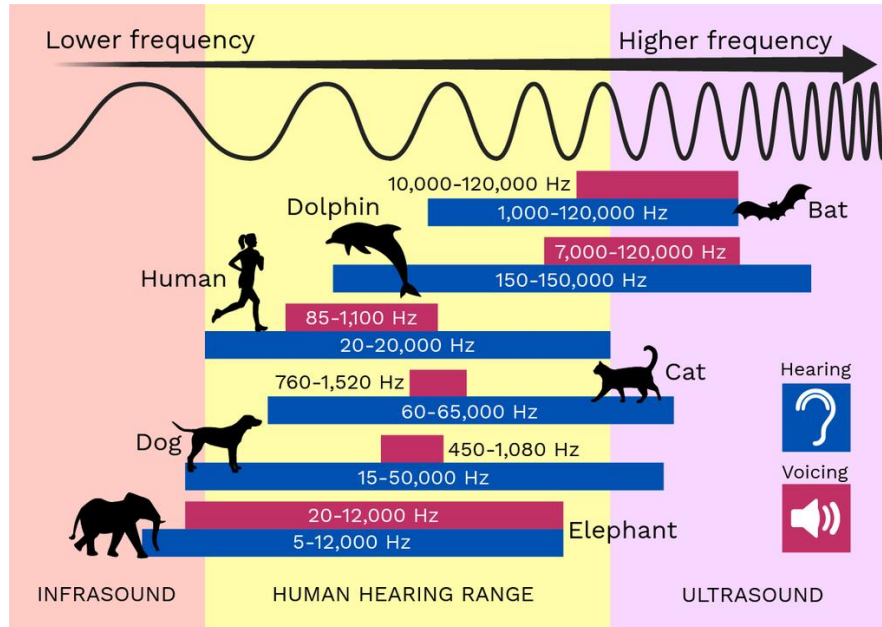


- This is called a Fourier Transform (FT)
- Therefore, we can restrict ourselves to sampling and reconstructing **sinusoidal signals**

How Fast Should We Sample?

What do you think the sampling frequency is for an audio CD?

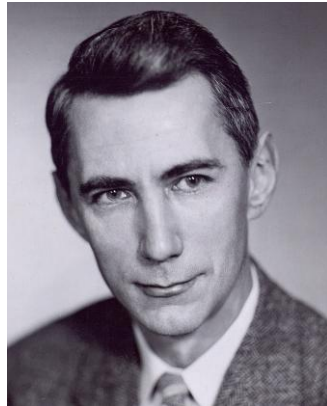
For context:



Typically 44,100 Hz

How Fast Should We Sample?

- In order to sample and later reconstruct a sinusoidal signal, the sampling frequency, f_s , must be **at least twice** the signal frequency, f
- In order to correctly sample a signal, f_s must be larger than $2 \times f_{max}$, where f_{max} is the largest frequency in the spectrum of the signal
 - f_{max} is called the Nyquist Frequency
- Based on **Nyquist-Shannon sampling theorem**



Influence of Sampling & Quantization

Example: Let's change up the sampling and quantization for the UM Fight Song

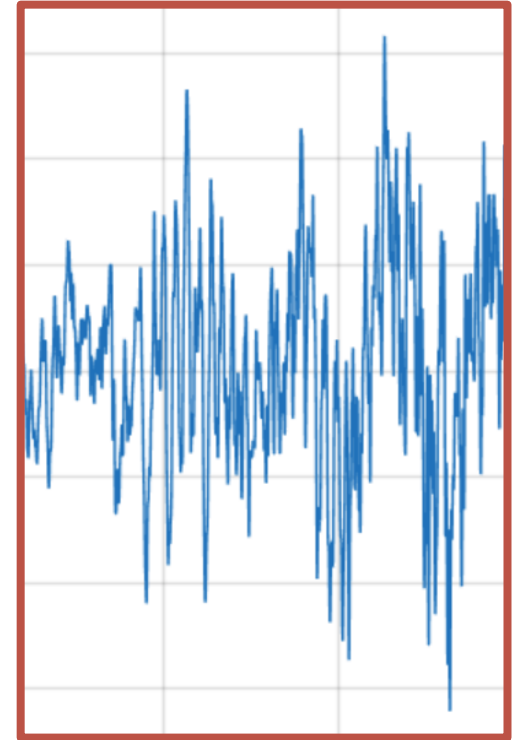
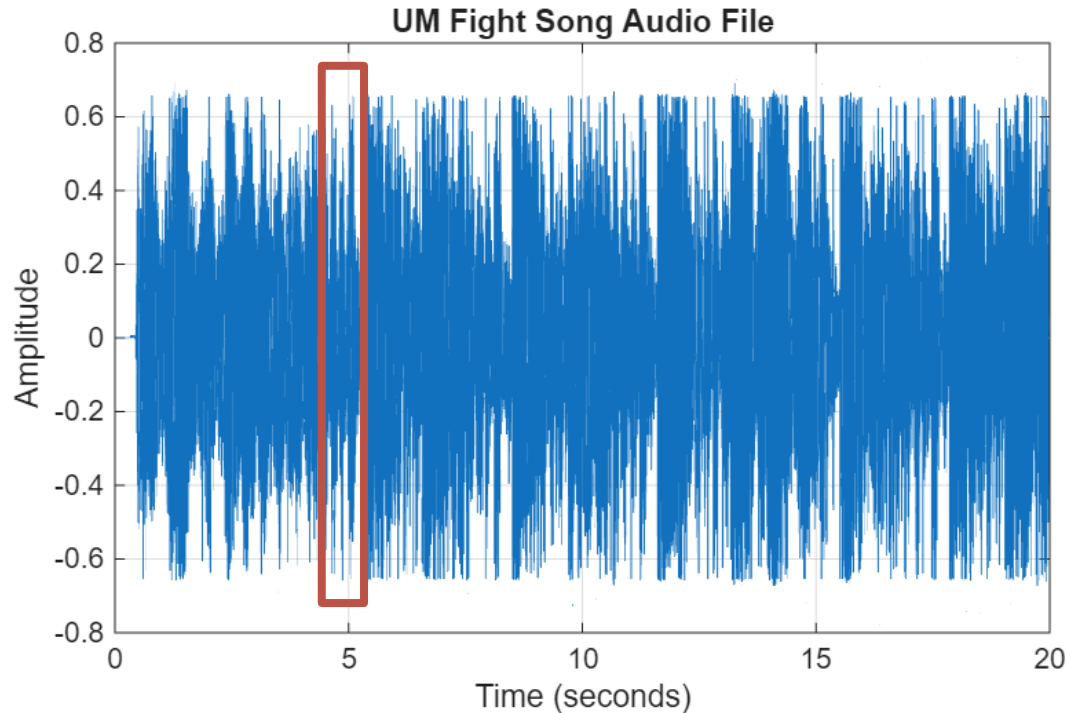
	32 Bit	8 Bit	4 Bit
44 kHz			
8 kHz			
1 kHz			

Music with lower sampling frequency **loses high-pitched sound**

Music with fewer discrete values **gets “noisy” (i.e. lots of static)**

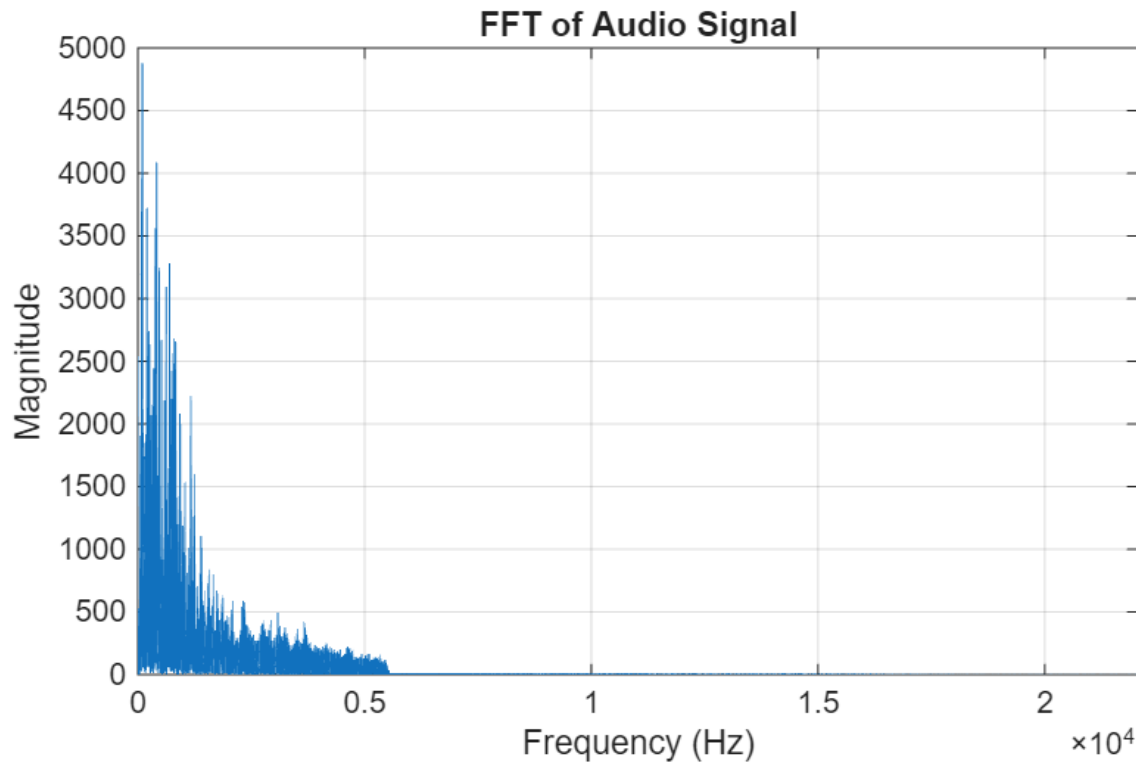
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Example: Let's change up the sampling and quantization for the UM Fight Song

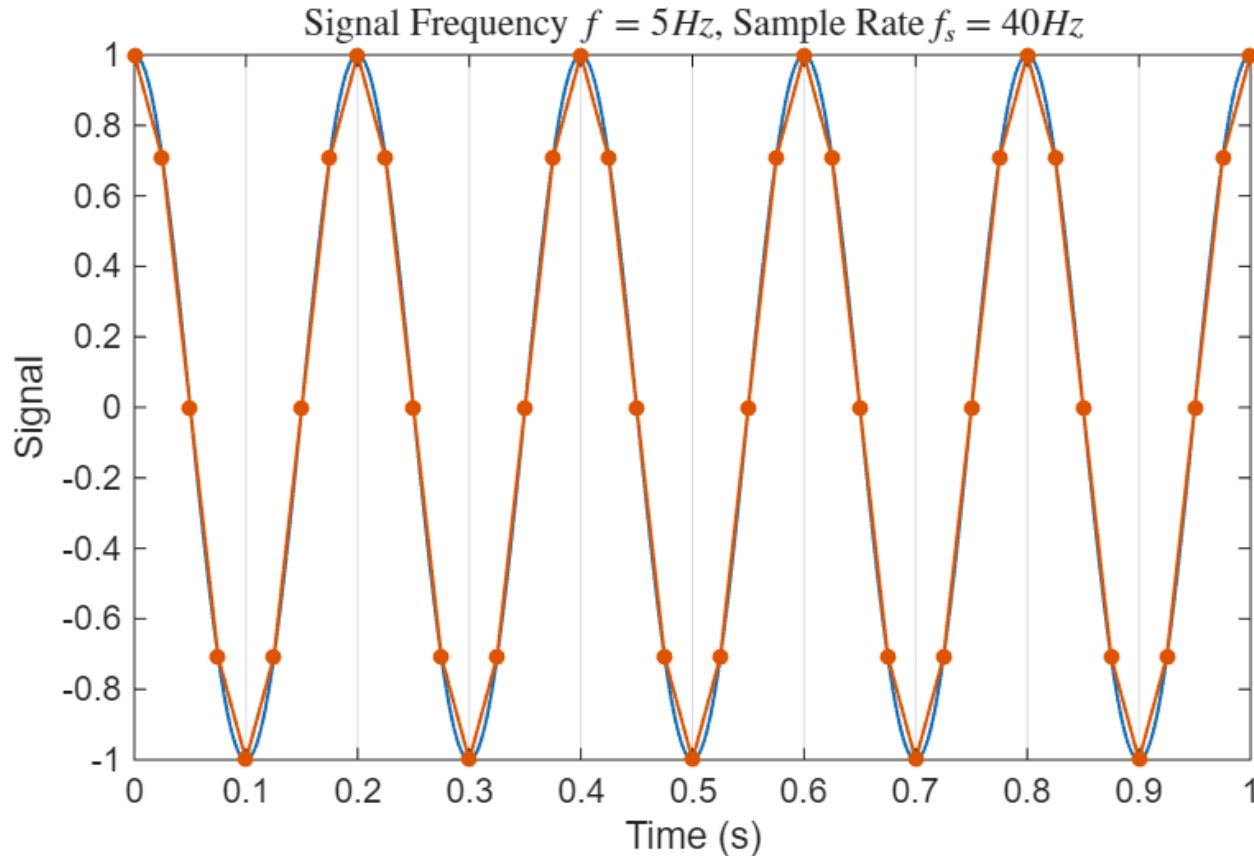


Influence of Sampling & Quantization

Example: Let's change up the sampling and quantization for the UM Fight Song

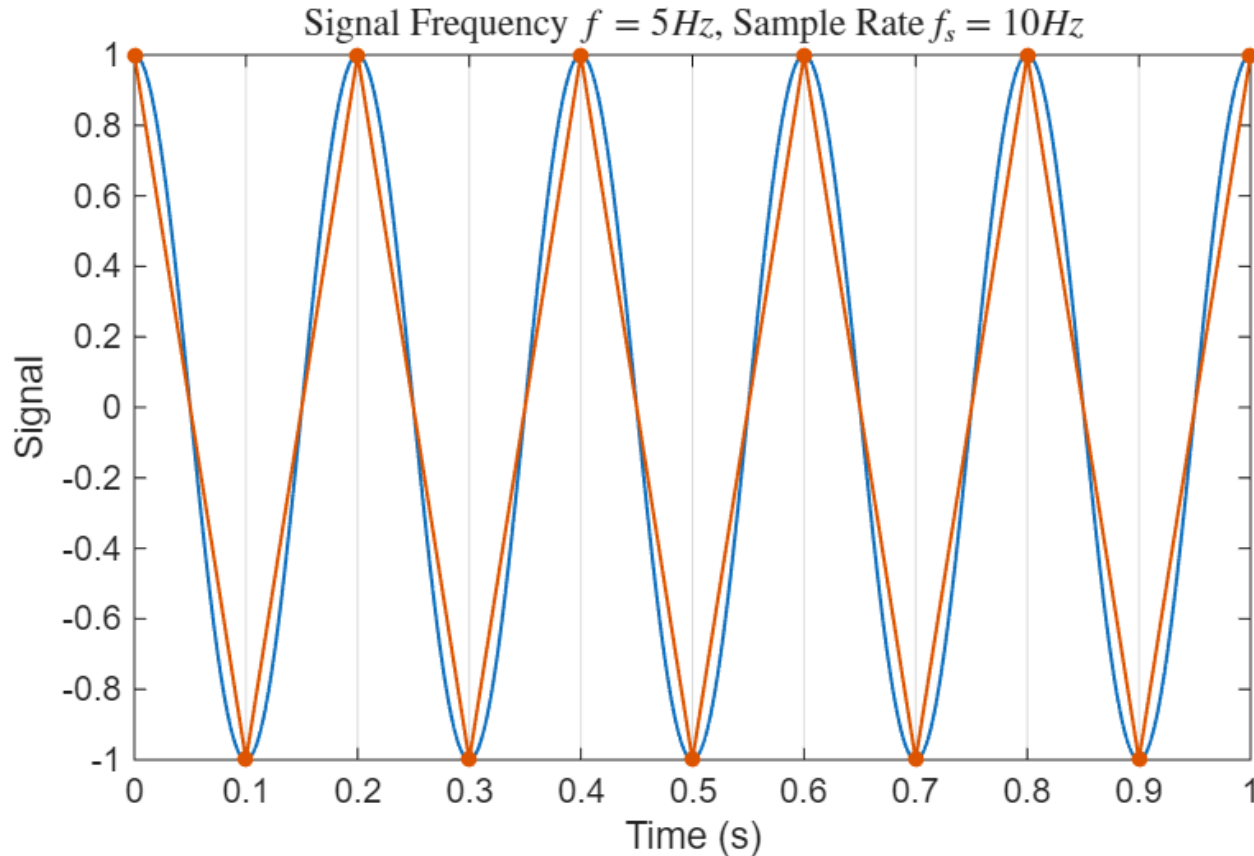


Sampling Frequency Effects - Example



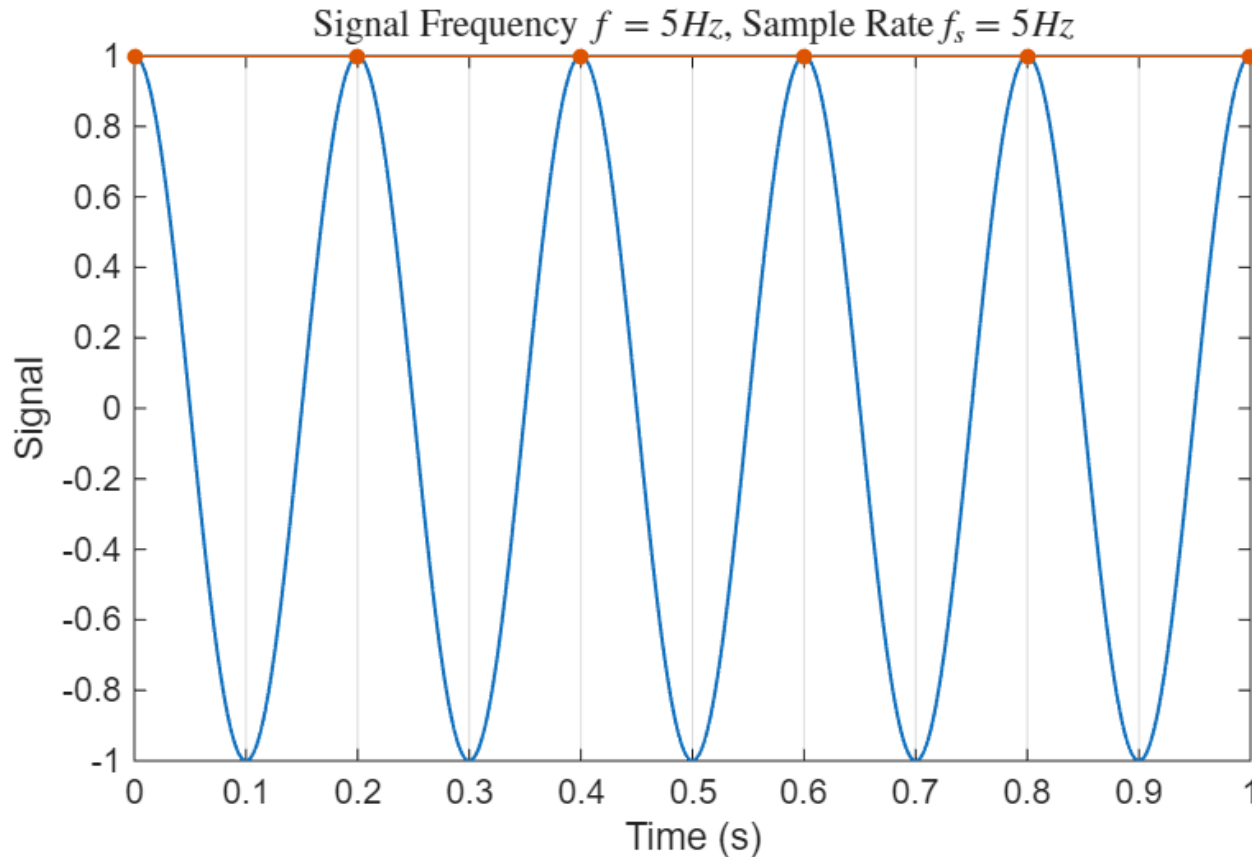
Signal can be
represented accurately

Sampling Frequency Effects - Example



Signal can be
represented accurately

Sampling Frequency Effects - Example

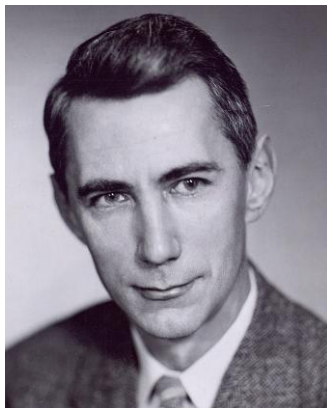


Signal **cannot** be represented accurately

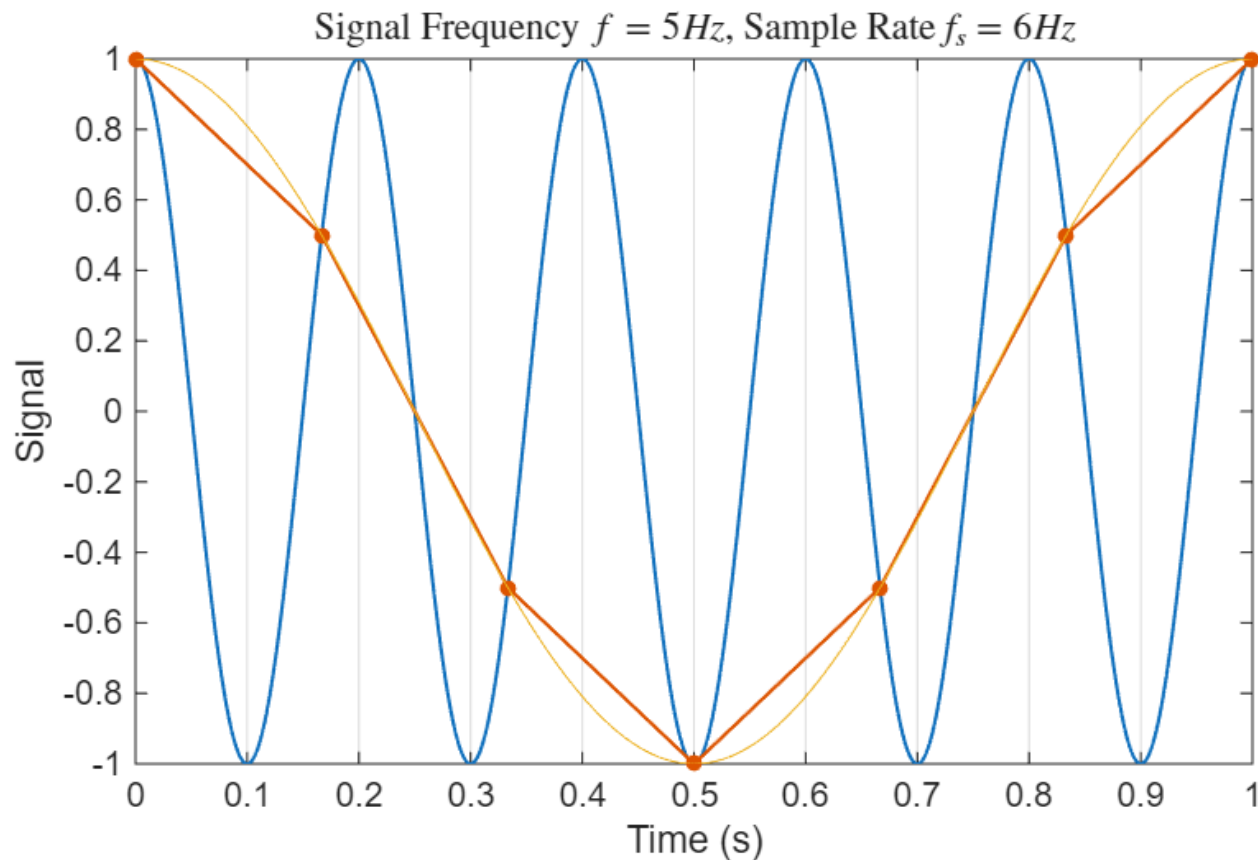
Recall the Nyquist-Shannon Sampling Theorem



- In order to sample and later reconstruct a sinusoidal signal, the sampling frequency, f_s , must be **at least twice** the signal frequency, f
- In order to correctly sample a signal, f_s must be larger than $2 \times f_{max}$, where f_{max} is the largest frequency in the spectrum of the signal
 - f_{max} is called the Nyquist Frequency
- Based on **Nyquist-Shannon sampling theorem**



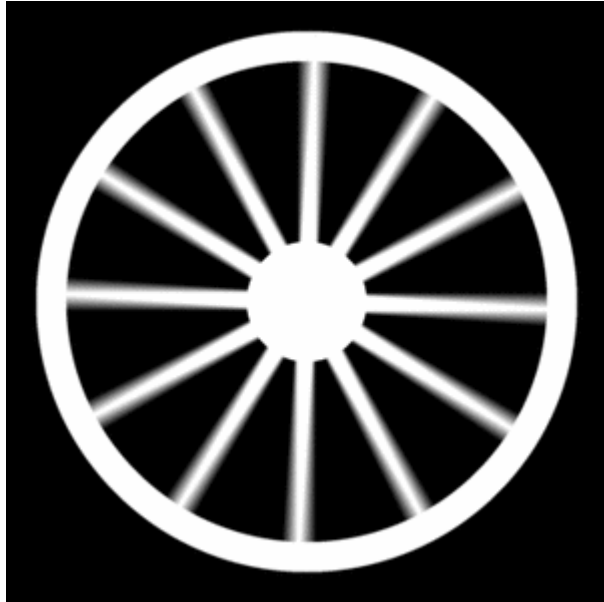
Sampling Too Slowly Leads to Aliasing!



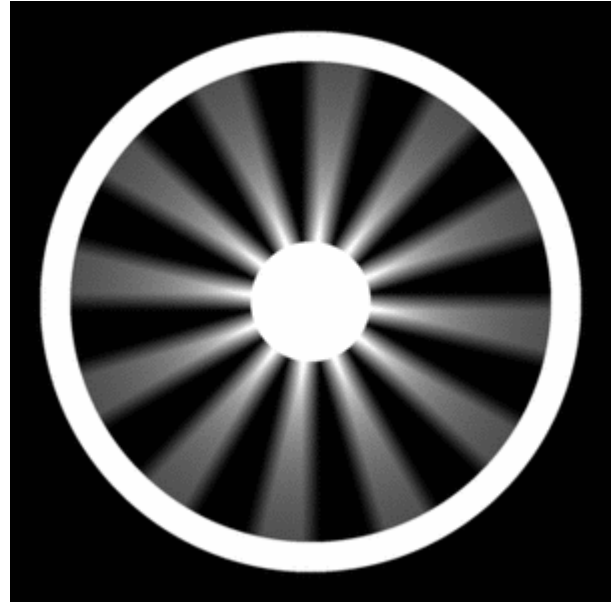
Signal **cannot** be represented accurately

Looks like a 1 Hz Signal!

Sampling Too Slowly Leads to Aliasing!



A wagon wheel rotating clockwise filmed at 24 frames per second



When spun clockwise faster, the wheel begins to look like it is rotating backward (called the wagon wheel effect)

- If a signal with a frequency larger than $0.5 \cdot f_s$ (where f_s is the sample frequency) is sampled, it will **alias**
 - It will appear as a **“phony”** signal of a lower **frequency**
- The aliased frequency (f_a) is given by

$$f_a = |n \cdot f_s - f|$$

where n is a positive integer and $f_a < 0.5f_s$

Aliasing Example

Calculate the aliased frequency for each of the following values of f using the formula $f_a = |n \cdot f_s - f|$, where n is a positive integer, $f_a < 0.5f_s$, and our sampling frequency $f_s = 10$ Hz

- $f = 8$ Hz

$$n=1: f_a = |1 \cdot 10 - 8| = 2 \text{ Hz}$$

Check if $f_a < 0.5f_s \dots 2 < 5 \checkmark$

- $f = 13$ Hz

$$n=1: f_a = |1 \cdot 10 - 13| = 3 \text{ Hz} \checkmark$$

$\uparrow 3 < 5$

Aliasing Example



Calculate the aliased frequency for each of the following values of f using the formula $f_a = |n \cdot f_s - f|$, where n is a positive integer, $f_a < 0.5f_s$, and our sampling frequency $f_s = 10$ Hz

- $f = 21$ Hz

$$n=1: f_a = |1 \cdot 10 - 21| = 11 \text{ Hz} \quad 11 \not< 5$$

$$n=2: f_a = |2 \cdot 10 - 21| = \boxed{1 \text{ Hz}} \quad 1 < 5 \checkmark$$

Aliasing Example

Calculate the aliased frequency for each of the following values of f using the formula $f_a = |n \cdot f_s - f|$, where n is a positive integer, $f_a < 0.5f_s$, and our sampling frequency $f_s = 10$ Hz

- $f = 27$ Hz

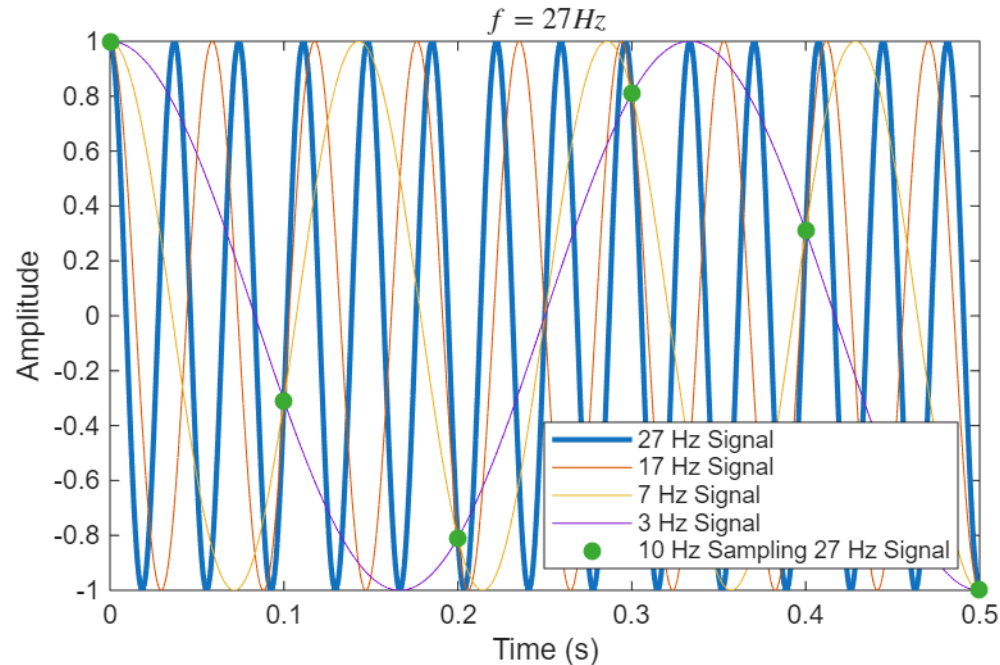
$$n=1: |1 \times 10 - 27| = 17 \text{ Hz} \rightarrow 17 \not< 5$$

$$n=2: |2 \times 10 - 27| = 7 \text{ Hz} \rightarrow 7 \not< 5$$

$$n=3: |3 \times 10 - 27| = \boxed{3 \text{ Hz}} \rightarrow 3 < 5 \checkmark$$

What is the meaning of the invalid solutions?

They are also signals that would intercept these sampling points, but we don't have enough data points to reconstruct them fully so we say they're invalid as we wouldn't arrive at them with only the sampled data, we would arrive at the smallest frequency

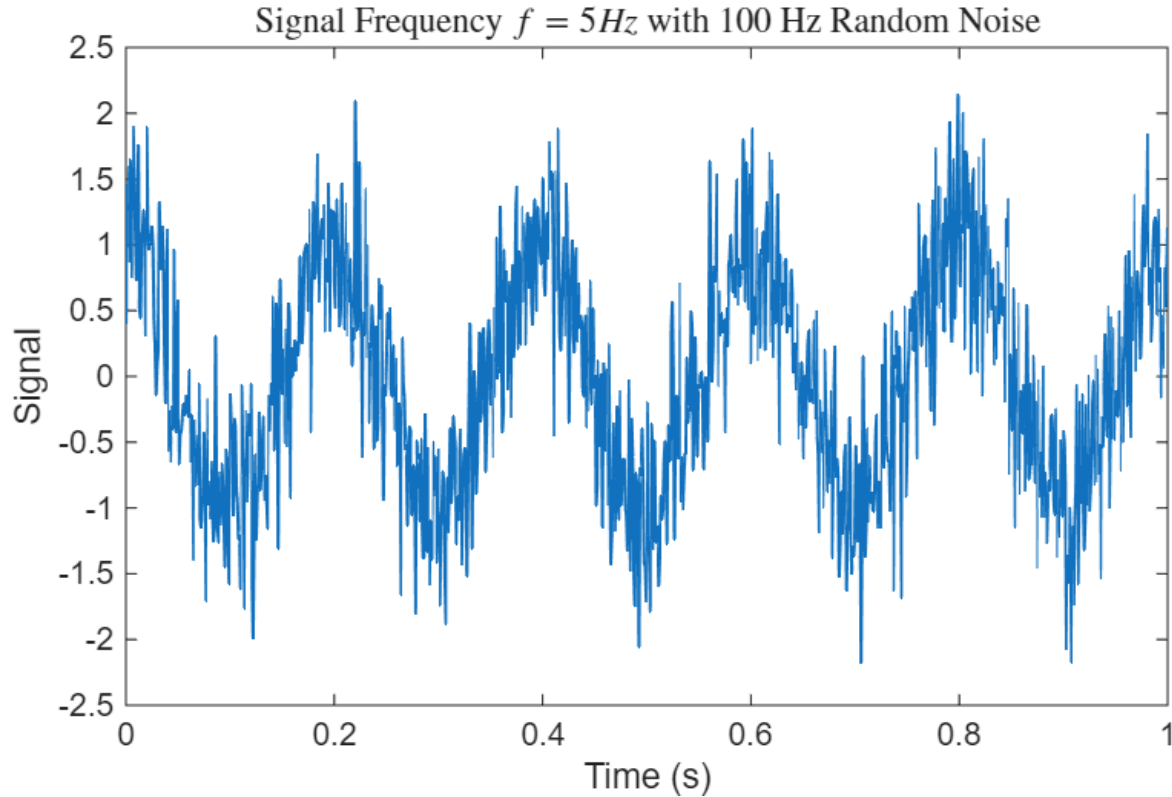


How is this Used in Practice?

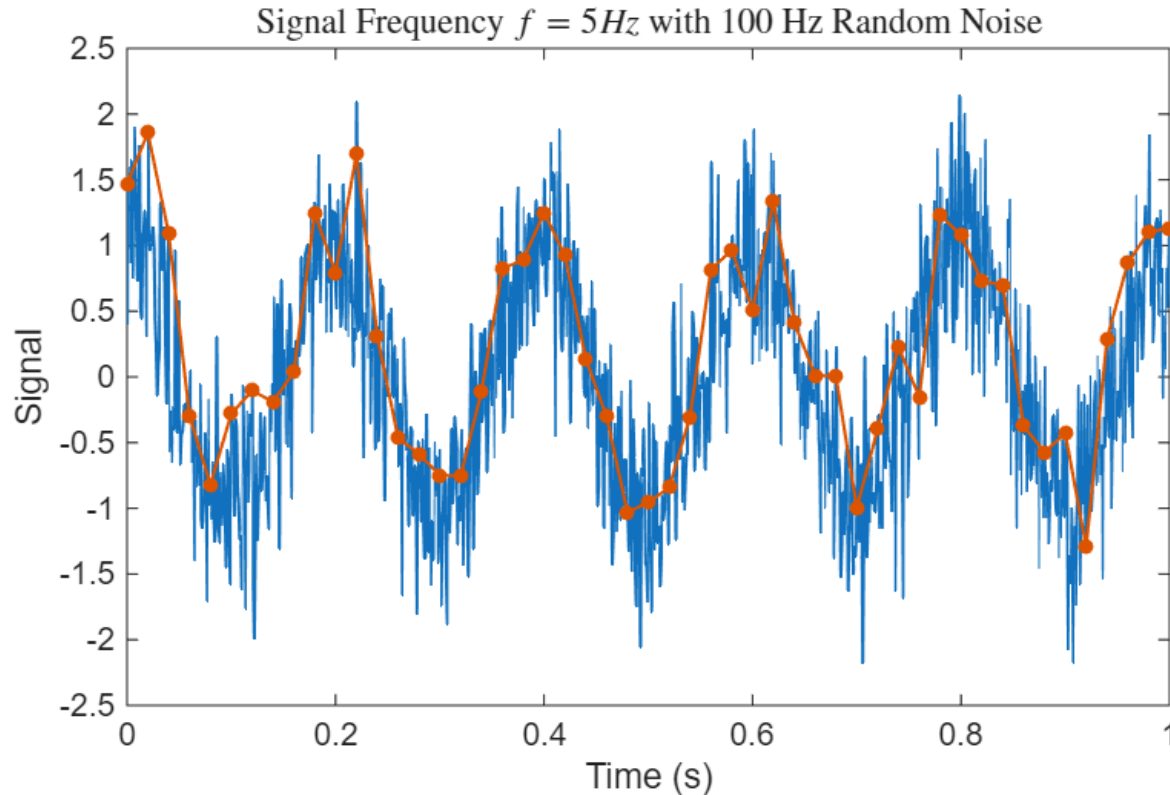


- Make f_s larger than twice the largest **useful signal** frequency in your input.
 - No need to oversample!
- Remove useless signals (i.e. noise) at frequencies higher than the Nyquist frequency (i.e. $f_s/2$) using **a low-pass filter** (aka anti-aliasing filter).
 - Without the low-pass filter your signal can be aliased and corrupt your useful signal

Example: 5 Hz Signal with Noise



Example: 5 Hz Signal with Noise



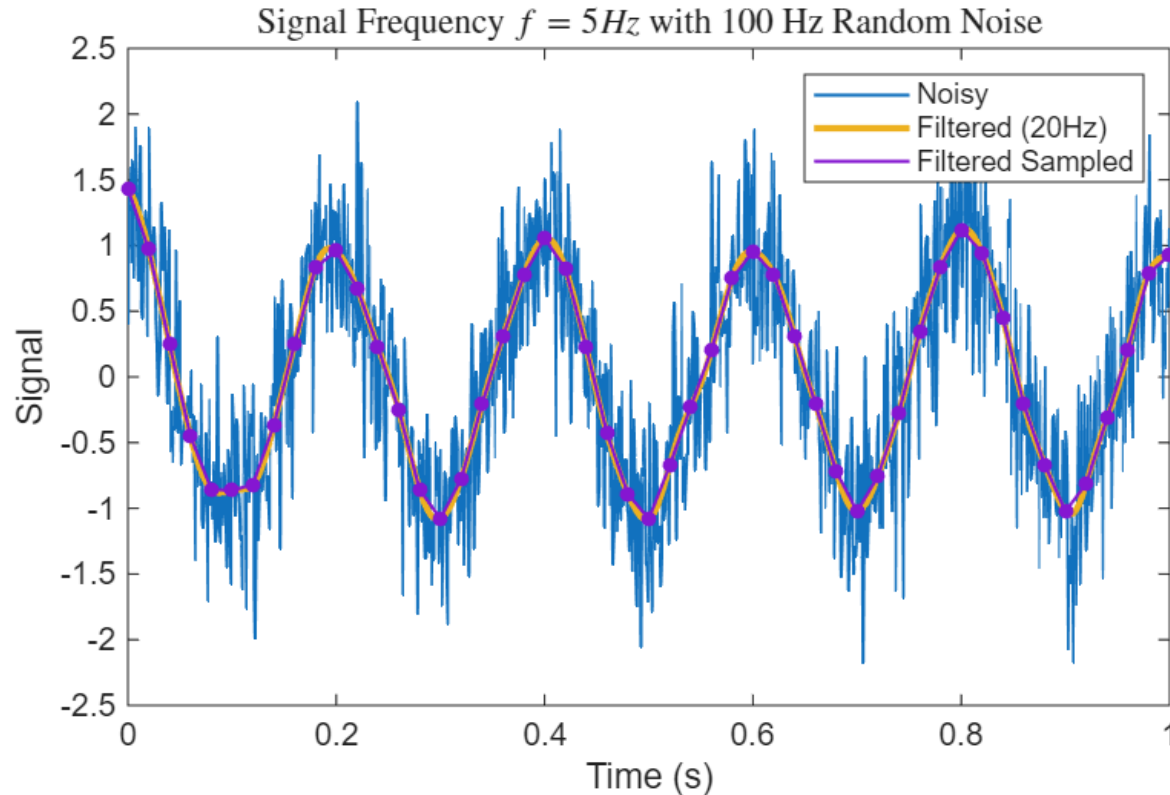
Sampling with $f_s = 50\text{ Hz}$ which is clearly greater than $2 \times f$

Noise gets aliased onto lower frequencies causing poor signal quality

Noise must be $>25\text{ Hz}$ ($f_s/2$)

We can use a low-pass filter with a cut-off frequency $\leq 25\text{ Hz}$ **before** sampling

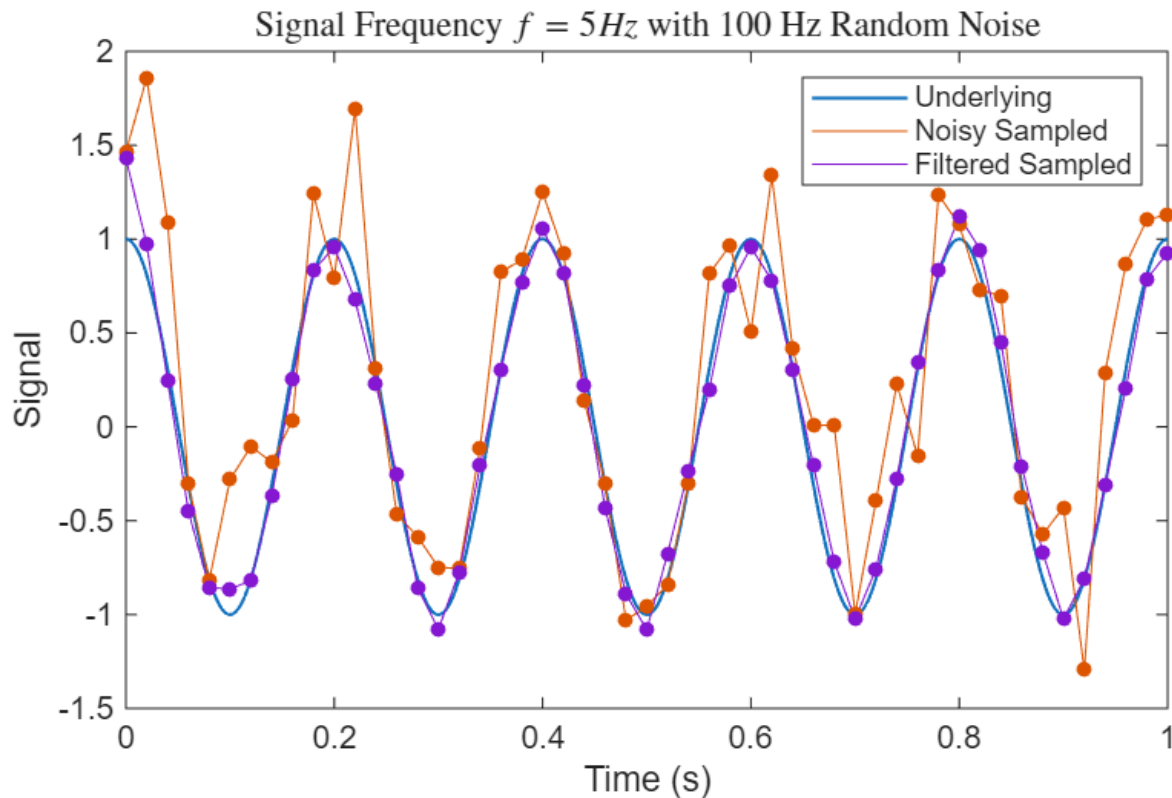
Example: 5 Hz Signal with Noise



Filtered using a low-pass filter with a 20 Hz cut-off

Now when we sample at 50 Hz we get a smoother result

Example: 5 Hz Signal with Noise



Moving Forward

- Let's now talk more about signals and ADC!

